Answer key: Lesson 7, Fill the Grid

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Each row has a circle, a square and a triangle. Which shape goes in the box with the ? mark in rows A, B and C?

Answer: circle; triangle; square

Row A has a triangle and a square, so the circle is missing. Row B has a circle and a square, so the triangle is missing. Row C has a triangle and a circle, so the square is missing.

Watch for: Choosing a shape that is already in the row.; Circling without looking at the other two shapes.

S2. Look at the three rows. Tick True or False for each statement.

Answer: A true; B false; C true

Row A has a square, a triangle and a circle: one of each, so True. Row B has two circles and no square, so False. Row C has a triangle, a circle and a square: one of each, so True.

Watch for: Saying True when a shape is used twice.; Looking at only the first two shapes.

S3. Every row and every column has a circle, a square and a triangle. Look at the bottom left box.

Answer: triangle; the column has a circle and a square

The column of the bottom left box has a circle at the top and a square in the middle, so the triangle is missing. The row has only a circle, so the row could not tell us.

Watch for: Choosing the row reason: the row has only one shape.; Choosing a shape that is already in the column.

S4. A child fills a grid. Row 1 is circle, square, triangle. Row 2 is square, triangle, circle. Row 3 is circle, square, triangle. Every row has one of each. Is the grid correct?

Answer: No

The first column is circle, square, circle: two circles. So the grid is not correct, even though every row has one of each.

Watch for: Saying Yes after checking only the rows.; Not building the grid to look at the columns.

S5. Every row of a 3 by 3 grid has one circle, and every column has one circle. How many circles are in the whole grid?

Answer: 3

There are 3 rows and each row has one circle, so there are 3 circles. (There are 3 columns with one circle each, which is the same 3 circles.)

Watch for: Writing 9, the number of boxes.; Writing 1, the circles in one row.

Think harder (page 3)

T1. Mi has a grid with only two shapes in it. Every row and every column has one of each. Which shape goes in each box?

Answer: triangle; circle; triangle

The top middle box is in the top row with a circle and above the square, so it cannot be a circle or a square: it is a triangle. Then the top row is circle, triangle, so the top right is a square. The middle row has a square, and its right box sits under a square, so the middle row is triangle, square, circle: the middle right box is a circle. The bottom row is square, circle, triangle, so the bottom right box is a triangle.

Watch for: Guessing a shape that is already in the row or the column.; Looking only along the row: the top middle box has two empty boxes in its row.

T2. Every row and every column of a grid has a circle, a square and a triangle. How many different shapes could go in each box?

Answer: 2; 1

First box: the circle is crossed out, so a square or a triangle could go in: 2. Second box: the triangle is crossed out by the row and the square by the column, so only the circle is left: 1.

Watch for: Counting the shapes that are already there instead of the shapes that could go in.; Forgetting that the row and the column can show the same shape.

T3. A grid is correct: every row and every column has one of each. A child changes it. Test each change with your tiles. Tick True or False.

Answer: first true; second false

Swapping two whole rows keeps one of each in every row, and every column still holds the same three shapes, so True. Swapping two boxes inside one row keeps that row correct, but the columns change and one column then has a repeat, so False.

Watch for: Checking only the rows after a swap.; Thinking that any swap breaks the grid.

Go further (page 4)

G1. Four by Four. Every row and every column has the digits 1, 2, 3 and 4 once each. Write the digits in the empty boxes.

Answer: 4, 1, 2, 1, 4, 3, 1 (reading the empty boxes row by row)

The grid is 1 4 2 3, 3 1 4 2, 2 3 1 4 and 4 2 3 1. For example, the third box of the bottom row: its column has 2, 4 and 1, so it is 3. The first box in the bottom row: its row has 2 and 3 and 1, so it is 4.

Watch for: Repeating a digit in a column.; Filling a box by looking only at the row when it has two empty boxes.

G2. Four by Four, with fewer clues. Only six digits are given. Every row and every column has 1, 2, 3 and 4 once each. Which digit goes in each box marked a, b and c?

Answer: 3; 4; 2

The grid is 2 1 4 3, 3 4 2 1, 4 3 1 2 and 1 2 3 4. For example, the first box of the third row: its row has 3 and 1, so it is 2 or 4; its column has 2 and 3 and 1 after we find them, so it is 4.

Watch for: Guessing a digit that is already in the row.; Stopping at the first digit that fits without checking the column.

G3. A 5 by 5 grid uses the digits 1, 2, 3, 4 and 5. Every row and every column has each digit once.

Answer: 5; 10

There are 5 rows and each row has one 3, so 5 threes. There are 5 fours and 5 fives, so 10 boxes hold a 4 or a 5.

Watch for: Writing 1 or 3 for the number of 3s.; Writing 4 or 5 for the boxes that hold a 4 or a 5.

Investigation (page 5): Grid builder

First: row 3 is 3, 1, 2. With a 2 in the middle there are exactly 4 grids: 1 3 2 / 3 2 1 / 2 1 3; 2 1 3 / 3 2 1 / 1 3 2; 2 3 1 / 1 2 3 / 3 1 2; and 3 1 2 / 1 2 3 / 2 3 1. Each is the same grid as another one turned upside down or swapped left to right, so a child who finds 2 can often find the other 2 by a swap.

Look for: A child who tries a corner, checks each row and column, and then swaps to get a new grid; a child who notices that a 2 must also be in the middle of the top and the bottom row of the corners; a child who says why there cannot be a fifth. Credit any clear reason. Do not mark a child down for stopping at one or two grids.

Exit ticket (page 5)

E1. A row has a circle and a square. Which shape is missing from the row? Circle one.

Answer: triangle

The row has a circle and a square. The triangle is missing, so the triangle goes in the box.

Watch for: Choosing a shape that is already in the row.; Circling without looking at the row.

E2. Which row has one of each? Choose the correct answer. Write its number in the bracket.

Answer: (2) square, triangle, circle

Row (2) has a square, a triangle and a circle: one of each. Row (1) has two circles, row (3) has two triangles and row (4) has two squares.

Watch for: Choosing (1) because it starts with a circle.; Not checking every shape in the row.

E3. A box has a circle in its row and a square in its column. Which shape goes in the box? Circle one.

Answer: triangle

Box a cannot be a circle, because the row has a circle. It cannot be a square, because the column has a square. The triangle is left.

Watch for: Choosing the circle or the square.; Looking at the row only.

Sneak-peek puzzles (not in the workbook)

1. circle
2. A square. The first box in the bottom row has a triangle and a circle above it in its column, so it must be a square. The row has only a triangle, so the column decides.